

CO2 Path

^^ first thing i want to have standard sizing throughout, ideally just want one tube size

^^ first gonna see what size the bigger components come in and try pick the same

CO2 Gas Cost

according to Air Products 2026 price list:

about: R1800

	K cylinder (~31kg, Inst/HP)	F cylinder (smaller, Inst/HP)
Gas fill	31kg × R36.01/kg = R1,116	est. 6-10kg × R193.25/kg = R1,160-1,933
Cylinder rental	R109.61	R109.61
Delivery charge	R152.75	R152.75
Admin charge	R107.00	R107.00
Subtotal excl VAT	R1,486	R1,529-2,302
All-in incl 15% VAT	≈ R1,708	≈ R1,758-2,647

50 000ppm CO2 sensor

1. R2162 only reads up to 50000ppm so won't know if overshoot (only reads up to 5%)
<https://www.robotics.org.za/sensors/co2-sensors/SEN0220>
2. R3467 - this CO2 sensor reads up to 100% concentration apparently - <https://za.rs-online.com/web/p/environmental-sensor-ics/1809026>

Pressure Regulator

GCE Pressure Regulator Co2, Ar 5/8 in G BSP - R1950 - https://za.rs-online.com/web/p/welding-regulators/0340129?cm_mmc=ZA-PLA-DS3A_-google_-PLA_ZA_EN_PMAX_Catch+All_-_-340129&matchtype=&&gclid=aw.ds&gad_source=1&gad_campaignid=20574379828&gbraid=0AAAAADkeWNOlpeVHm_rzm8CDnY9KSQMeG&gclid=Cj0KCQjw16_UBhCqARIsAldOaXyXcNBWxcqel5a6kFxR31l1uT1bUOnmX5QQpJ1C6nZ7MopZoFXdP6AaAq18EALw_wcB

Solenoid Valve

SMC Solenoid Valve - NC 24V - R1037 - https://za.rs-online.com/web/p/pneumatic-solenoid-valves/0425481?cm_mmc=ZA-PLA-DS3A_-google_-PLA_ZA_EN_PMAX_Catch+All_-_-425481&matchtype=&&gclid=aw.ds&gad_source=1&gad_campaignid=20574379828&gbraid=0AAAAADkeWNOlpeVHm_rzm8CDnY9KSQMeG&gclid=Cj0KCQjw16_UBhCqARIsAldOaXyXcNBWxcqel5a6kFxR31l1uT1bUOnmX5QQpJ1C6nZ7MopZoFXdP6AaAq18EALw_wcB

Flow control/needle valve

Check valve

^^^ Option that Includes: Regulator, solenoid, needle valve, check valve

- regulator, solenoid, fine tune valve, check valve, Connector: G5/8 : R1628, <https://mcmerwe.co.za/shop/lst-aqua-co2-dual-gauge-regulator/>, would rather go for this because it includes a bunch of components I need and in one unit that is proved to work together - it's less expensive and less risky and It is local so I can go fetch it myself and talk with the vendors to clarify it will work

Tubing

- for mcmerwe one - tube R75: <https://mcmerwe.co.za/shop/aquapro-co2-pressure-resistant-tube-3m/>

Compression fitting / bulkhead connector

Bulkhead Connector	RS PRO PMM Series Pneumatic Fitting, 6 mm Push-in, 6 mm Push-in	RS Online	2	R134	R268		https://za.rs-online-2351214&match=TUBhAiEiwADv8c
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